

LIKITHA TADAKALA

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PROFESSIONAL SUMMARY

Engineer with 2+ years of production experience across Machine Learning, Data Science, and Software Engineering; shipped YOLO and VLM pipelines at 90 to 95 percent mAP and a data cleaner 63x faster than T5. AWS Certified. Strong in Python, AWS, SQL, Power BI.

TECHNICAL SKILLS

Languages and Tools: Python, SQL, R, JavaScript, Git, GitHub, Docker, Linux, Bash, REST APIs

ML and Deep Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Transfer Learning, Fine Tuning, Model Evaluation, Feature Engineering, Explainable AI

LLMs and Generative AI: RAG, Vision Language Models, LangChain, Ollama, Mistral, LLaMA2, Google Gemini, FAISS, Prompt Engineering, Offline Inference

Cloud and MLOps: AWS (SageMaker, Lambda, S3, EC2, IAM, Rekognition, Textract), FastAPI, Flask, Model Deployment, Batch Inference, CI/CD, Model Monitoring

Data and Analytics: Pandas, NumPy, EDA, Statistical Modeling, Predictive Modeling, Power BI (DAX), Tableau, Matplotlib, Seaborn

Computer Vision and Certs: YOLO, DETR, OpenCV, Object Detection, Image Segmentation, OCR

PROFESSIONAL EXPERIENCE

Actual Reality Technologies | Data Scientist | Perrysburg, OH

Sep 2025 to Present

- Built and shipped an end to end ML inference pipeline integrating YOLOv11-X, DETR, Google Gemini 2.5 Flash VLM, and AWS Textract OCR on 700+ engineering drawings at 2560px, hitting **90 to 95 percent mAP@0.5:0.95**, eliminating 30 to 40 percent rebar weight estimation errors, and cutting manual processing time 70 percent.
- Orchestrated AWS ML infrastructure for scalable batch inference; resolved 37 to 1.7 percent class imbalance via synthetic data generation; built Power BI dashboards for KPI monitoring; co-authored arXiv paper.

CodesOnBytes | Jr Data Scientist | Remote

Dec 2023 to Jan 2024

- Built a **Python** (Pandas) data ingestion pipeline to harvest and clean raw data from public REST APIs; developed a **linear regression forecasting model** in scikit-learn and conducted EDA with Matplotlib and Seaborn to identify outliers and feature correlations before model training.

Ozibook | Data Analyst | Remote

Mar 2023 to Dec 2023

- Executed complex **SQL queries** across operational workflows to identify process bottlenecks, supporting data driven product and business decisions.
- Designed and deployed interactive **Power BI dashboards** visualizing key business metrics, reducing weekly reporting cycle time and giving leadership real time KPI visibility; partnered with stakeholders to define metrics and validate data quality.

Intel Corporation | Industrial Trainee | Remote

Apr 2023 to Jul 2023

- Fine tuned **YOLOv5 in PyTorch** on a Roboflow automotive dataset, achieving **90 percent mAP@0.5** across 3 classes; applied focal loss and augmentation to resolve class imbalance and calibrated the accuracy and latency trade off for real time inference.

PROJECTS

SmartClean AI | Hybrid ML Framework for Automated Data Cleaning

- 63x faster than **Hugging Face T5** (33ms vs 2s per 1,000 records), 100 percent format consistency vs T5's 90.3 percent, and 98 percent quality score; deployed as a **Flask REST API** with zero API costs.

YouTube Learning Chatbot | Privacy Preserving RAG Pipeline

- Built an offline **RAG system** using FastAPI, Gradio, FAISS, and locally hosted LLMs (Mistral, LLaMA2 via Ollama) with transcript chunking for unstructured video content and zero external data transmission.

Vision For The Blind | Edge AI Assistive System on Raspberry Pi 4

- Deployed **Inception V3** (97.3 percent accuracy, 124 classes via transfer learning) and **Tesseract OCR** on Raspberry Pi 4 with a frame deletion memory pipeline for sub second audio feedback on edge hardware.

PUBLICATIONS AND CERTIFICATIONS

- Beyond Visualization: Building Decision Intelligence Through Iterative Dashboard Refinement, arXiv, 2025.
- VISION: Visual Insight Solution Interface Outreach Navigator, IEEE Xplore.
- AWS Certified AI Practitioner, AWS Certified Cloud Practitioner

EDUCATION

University of Toledo | MS Computer Science (AI) Aug 2024 to May 2026 | GPA: 4.0 / 4.0

Malla Reddy College of Engineering | B.Tech CSE (Data Science) Aug 2020 to May 2024 | GPA: 3.6 / 4.0